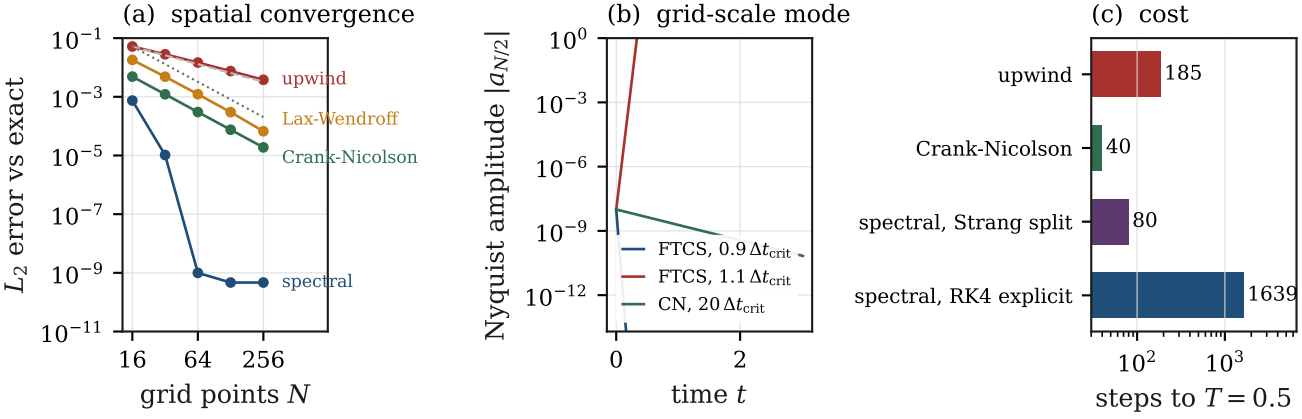
(a) spatial convergence
L_2 error vs exact
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
upwind
Lax-Wendroff
Crank-Nicolson
spectral
16
64
256
grid points N
(b) grid-scale mode
Nyquist amplitude $|a_{N/2}|$
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
FTCS, $0.9\,\Delta t_{\mathrm{crit}}$
FTCS, $1.1\,\Delta t_{\mathrm{crit}}$
CN, $20\,\Delta t_{\mathrm{crit}}$
0
2
time t
(c) cost
upwind
185
Crank-Nicolson
40
spectral, Strang split
80
spectral, RK4 explicit
1639
10^{2}
10^{3}
steps to $T = 0.5$